

MD JUBIER AHAMMED

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EDUCATION

M.S. in Geography and Environmental Studies, Texas Tech University

Aug 2025 – Expected May 2027

Lubbock, TX | Current GPA: 3.863 (as of Summer 2026)

Thesis: working title to be determined. Plans to apply a Landsat random forest classification framework in Google Earth Engine to coastal land cover in the Mississippi Delta, in parallel with an ongoing wetland-mapping collaboration with Dr. Chao Xu in the Ganges Delta.

B.S.S. in Disaster and Human Security Management, Bangladesh University of Professionals

Jan 2020 – Jan 2024

Dhaka, Bangladesh | GPA: 3.90/4.00

Undergraduate thesis: Assessing and Prediction of Spatio-temporal Changes of LULC using CA-ANN Model and its Socioeconomic Implications for Coastal Bangladesh: A Case Study of Dacope Sub-district

RESEARCH INTERESTS

Natural hazards and disaster risk; remote sensing and GIS; machine learning and GeoAI; spatial analysis; social vulnerability and climate adaptation; land use / land cover change; coastal wetland mapping.

PUBLICATIONS

Under Review

Ahammed, M.J., Xu, C.*, Xu, Y., Zhou, H. Spatial Pattern of Post-storm Recovery Revealed by Nighttime Light: Evidence from Harris County, Texas. Under review, ISPRS International Journal of Geo-Information. Preprint: <https://doi.org/10.20944/preprints202609.0372.v1>

Ahammed, M.J.*, Sonet, M.S. Social Vulnerability and Compound Heat-Flood Exposure in Houston, Miami, and Norfolk: A Census-Tract-Level Analysis. Under review, Regional Environmental Change. Preprint: <https://doi.org/10.21203/rs.3.rs-10349570/v1>

**Corresponding author*

In Preparation

Xu, C., Ahammed, M.J. Tracking Coastal Wetland Dynamics in the Ganges Delta from 1985–2026 Using Google Earth Engine. In preparation.

Conference Presentations

Ahammed, M.J. Dynamic Trends in Land Use and Land Cover Change with QADI Validation: Exploring the Shifts in Shrimp Farming and Socioeconomic Implications in Coastal Bangladesh's Dacope Sub-district. International Conference on Environmental Science and Resource Management (ICESRM). Presented; no online proceedings available.

RESEARCH EXPERIENCE

Graduate Research, Department of Geosciences, Texas Tech University

Aug 2025 – Present

Supervisor: Dr. Chao Xu

- Conducted a spatial and quantile analysis of infrastructure precarity and social vulnerability during Winter Storm Uri in Harris County, using VIIRS Black Marble nighttime light deficit data against ACS and social vulnerability variables; methods included Spearman correlation, OLS with clustered standard errors, robust regression, quantile regression, Moran's I, and a spatial lag model.
- Developing a Landsat random forest classification of tidal flats and wetlands in the Ganges Delta in Google Earth Engine, using manually collected ground-truth points and a revised land-cover scheme.

Independent Research Collaboration

2025 – Present

With M. Shahriar Sonet (Ph.D. candidate, Geospatial Information Sciences, University of Texas at Dallas)

- Conducting a census-tract-level analysis of compound heat-flood exposure and social vulnerability across Houston, Miami, and Norfolk, integrating NFIP flood claims, CDC social vulnerability data, NHGIS tract crosswalks, and GEE-derived land surface temperature.

Research Consultant, Remote Sensing Division, CEGIS

Jan 2024 – Dec 2024

Center for Environmental and Geographic Information Services, Dhaka, Bangladesh

- Created, updated, and digitized land use / land cover maps for Chittagong division.
- Georeferenced satellite imagery for digitization.
- Managed databases and resolved topological errors.
- Conducted crop pattern analysis.

Research Assistant, Geospatial Health Studies Division, CHIRAL

Apr 2023 – Dec 2023

Center for Health Innovation, Research, Action and Learning – Bangladesh

- Collected spatial and non-spatial data and conducted thematic analysis on environmental factors influencing dengue outbreaks, using GIS for disease mapping and risk assessment.
- Trained interns on mobile data collection and GIS tools (KoboToolbox, ArcGIS).
- Analyzed and visualized data in Excel and SPSS, producing charts and infographics for project reporting.

CLASS PROJECTS

Site Suitability Analysis for a Bison Preserve (“Buffalo Commons”), GIST 5300*Multi-criteria weighted suitability model in ArcGIS Pro across ten Great Plains states.***Spatial Database for EV Charging Deserts in Texas, GIST 5304***PostgreSQL/PostGIS spatial database and spatial SQL queries relating to charger locations, highways, and census tracts.***Hedonic Rent Pricing Model, Lubbock, Texas, GEOG 5309***OLS regression in R modeling monthly rent against unit attributes and neighborhood income, with diagnostics.***TEACHING EXPERIENCE**

Graduate Teaching Assistant, Department of Geosciences, Texas Tech University

Aug 2025 – Present

Courses: Introduction to GIS; Regional Geography of the World; Global Environmental Science

- Grade coursework and exams.
- Assist students during lab sessions.
- Hold weekly office hours.

AWARDS, FELLOWSHIPS & FUNDING

Graduate Teaching Assistantship, Texas Tech University (Aug 2025 – Present)

Department of Geosciences Scholarship, Texas Tech University (Fall 2026)

Graduate Field Experience Fellowship, Texas Tech University (Summer 2026)

Gary Elbow Scholarship, Department of Geosciences, Texas Tech University (Fall 2025)

TECHNICAL SKILLS

GIS & Remote Sensing: ArcGIS Pro, ArcMap, QGIS, Google Earth Engine, ENVI

Programming & Databases: Python, JavaScript (Google Earth Engine), R, SQL/PostGIS

Statistics: SPSS, Jamovi

Remote Sensing Data: Landsat, Sentinel, MODIS, VIIRS Black Marble

Data Collection: KoboToolbox

Graphics: Adobe Illustrator, Canva

LANGUAGES

Bengali — native

English — fluent (IELTS overall band 7.5)